PRE-ALGEBRA (7TH) · UNIT 9: SPATIAL THINKING

Skate Park Designer

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students design a skate park. Ramps and rails provide angle pairs and parallel lines; the fun box and deck use polygon angle sums; the bowl uses circle formulas; and the logo uses transformations and symmetry.

The design constraints (ramp angles) give students reasons to compute and justify angles.

Standards Addressed

- 7.G.B.5 Supplementary, complementary, vertical, and adjacent angles
- 8.G.A.5 Angles formed by parallel lines and a transversal; angle sums
- 7.G.B.4 Circumference and area of circles
- 8.G.A.1–3 Transformations

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with skate park photos. Part 1 ramp angles.
Day 2	Part 2 parallel rails and polygons.
Day 3	Part 3 bowl, logo, and blueprint.

Materials

- Student handout and Feature Cards
- Protractors and rulers
- Compasses
- Grid paper and tracing paper

How the Three Versions Differ

Level	What changes
CP2	Supplementary, complementary, vertical, and corresponding angles with numbers, hexagon sums, circumference, and a translation.
CP1	Algebraic angle equations, congruence, polygon sums, circle area and cost, and rotations.
Honors	Quadratic angle equation, proofs, tiling, annulus area, and composite transformations.

Watch For (Common Misconceptions)

- Confusing complementary (90°) with supplementary (180°).
- Using $(n - 2)180$ for each angle instead of the sum.
- Rotating clockwise instead of counterclockwise.

Extension Ideas

- Build a cardboard model of one feature.
- Visit or research a local skate park.
- Create a logo using a computer drawing tool.

Answer Key – CP2

Part 1: Ramp Angles

#	Task	Answer
1	The quarter pipe ramp angle is 30°. What is the supplementary angle on the other side?	180 − 30 = 150°
2	A support post makes a right angle with the ground. The ramp makes 30° with the ground. What angle does the ramp make with the post?	90 − 30 = 60° (complementary)
3	Two rails cross and make a 70° angle. What are the other three angles?	110°, 70°, 110° (vertical angles are equal)
4	Name one pair of complementary and one pair of supplementary angles you found.	30° and 60°; 30° and 150° (or 70° and 110°)

Part 2: Rails & Deck

#	Task	Answer
5	The path makes a 65° angle with the first rail. What is the corresponding angle at the second rail?	65°
6	What is the same-side interior angle to the 65° angle?	115°
7	The fun box top is a regular hexagon. Find the sum of its angles.	(6 − 2) × 180 = 720°
8	Find the measure of each angle of the regular hexagon.	720 ÷ 6 = 120°

Part 3: Bowl & Logo

#	Task	Answer
9	The bowl is a circle with diameter 24 ft. Find its circumference.	$C = \pi d \approx 3.14 \times 24 = 75.36$ ft
10	Translate the point (2, 3) on the logo 4 units right and 1 unit down.	(6, 2)
11	Draw a logo with at least one line of symmetry. Show the line.	Answers vary.

Reflection

#	Prompt	Look for
R1	How do parallel rails help you find angles without measuring?	Corresponding and alternate angles are equal, so one angle gives the others.

#	Prompt	Look for
R2	Why does a regular hexagon have 120° angles?	The 720° sum is shared equally by 6 angles.

Answer Key – CP1

Part 1: Ramp Geometry

#	Task	Answer
1	A ramp and its support form a right triangle with the ground. If the ramp angle is $(2x + 4)°$ and the angle at the top is $(3x + 1)°$, find x and both angles.	$2x + 4 + 3x + 1 = 90 \rightarrow x = 17$; angles 38° and 52°
2	Does the 38° ramp meet the rule (15° to 35°)? What angle would you change it to?	No; change it to 35° or less (e.g., 30°).
3	Two crossing rails form angles $(4y - 10)°$ and $(2y + 50)°$ that are vertical. Find y and the angle.	$4y - 10 = 2y + 50 \rightarrow y = 30$; angle 110°
4	Explain why vertical angles are always equal.	Each is supplementary to the same adjacent angle.

Part 2: Layout

#	Task	Answer
5	An alternate interior angle is $(5x - 20)°$ and its partner is $(3x + 30)°$. Find x and the angle.	$5x - 20 = 3x + 30 \rightarrow x = 25$; 105°
6	The fun box top is a regular hexagon and the pyramid base is a square. Find each interior angle and the sum of the exterior angles of each.	Hexagon 120°, square 90°; exterior sums are 360° for both
7	Two ramp triangles have matching sides 6 ft, 8 ft, 10 ft. Are they congruent? Which rule?	Yes, SSS
8	What polygon has an interior angle sum of 1,080°? Could it be the deck?	$(n - 2)180 = 1{,}080 \rightarrow n = 8$, an octagon; yes

Part 3: Bowl & Logo

#	Task	Answer
9	Find the circumference and area of the 24-ft bowl. A coping rail goes around the edge at $12 per foot. Find its cost.	$C \approx 75.36$ ft, $A \approx 452.16$ sq ft; cost $75.36 \times 12 \approx \$904.32$
10	The logo is a triangle with vertices (1, 1), (4, 1), (1, 3). Rotate it 90° counterclockwise about the origin. Give the new vertices.	(−1, 1), (−1, 4), (−3, 1)
11	Design a logo with rotational symmetry of order 3 or more. Give its angle of rotation.	Answers vary; e.g., order 3 → 120°.

Reflection

#	Prompt	Look for
R1	How did writing equations help you find angles?	Angle relationships (sum to 90°, 180°, or equal) become equations that can be solved.
R2	How is a rotation different from a translation?	A rotation turns a figure about a point; a translation slides it without turning.

Answer Key – Honors

Part 1: Angle Proofs

#	Task	Answer
1	A path crosses two parallel rails. One angle is $(x^2 - 10)°$ and its corresponding angle is $(3x + 30)°$ with $x > 0$. Find x and the angle.	$x^2 - 3x - 40 = 0 \rightarrow (x - 8)(x + 5) = 0 \rightarrow x = 8$; angle 54°
2	Prove that same-side interior angles are supplementary using a corresponding angle and a linear pair.	Corresponding angles are equal; the linear pair sums to 180°; substitute.
3	A ramp rises 4 ft for each 10 ft forward. Use a protractor drawing to estimate the angle. Does it meet the rule?	About 22°; yes
4	The angles of a triangular kicker are in the ratio 1 : 2 : 6. Find them. Does the smallest meet the ramp rule?	20°, 40°, 120°; yes (20° is between 15° and 35°)

Part 2: Deck Design

#	Task	Answer
5	Which regular polygons tile a flat deck by themselves? Use interior angles to explain.	Triangles (60°), squares (90°), hexagons (120°); their angles divide 360°.
6	Can regular octagons and squares tile together? Show the angle sum at a vertex.	Yes: 135 + 135 + 90 = 360
7	A regular polygon deck has interior angles of 150°. How many sides? What is its angle sum?	Exterior 30° → 12 sides; sum 1,800°
8	Explain why every regular polygon has exterior angles that add to 360°.	Walking around, you turn a full circle; each exterior angle is one turn.

Part 3: Bowl & Logo

#	Task	Answer
9	The bowl is a circle with diameter 24 ft. A safety zone 4 ft wide surrounds it. Find the area of the safety zone in terms of π and to the nearest square foot.	$\pi(16^2 - 12^2) = 112\pi \approx 352$ sq ft
10	Reflect the logo triangle (1, 1), (4, 1), (1, 3) across the y-axis, then rotate the image 180° about the origin. What single transformation does the same thing?	Reflect: (−1, 1), (−4, 1), (−1, 3). Rotate 180°: (1, −1), (4, −1), (1, −3). Same as a reflection across the x-axis.
11	Design a logo with exactly 4 lines of symmetry and rotational symmetry of order 4. Explain how you know.	Answers vary; e.g., a square-based design; rotations of 90° map it onto itself.

Reflection

#	Prompt	Look for
R1	Why do only some regular polygons tile a plane?	The interior angles at a vertex must add to exactly 360°.
R2	How can two transformations combine to equal a single transformation?	The composite maps every point the same way; here, a y-axis reflection then a 180° rotation equals an x-axis reflection.